Group 1 · Ship Design Project 2026

11 November 2026 · T. Santos, L. Batista

Ship and Ocean Systems Design 2026 · Ship Design Project · Scope sheet for your group

Field	Value
Group	Group 1
Members (student number)	106238 · 107304 · 106645
Baseline	B · Azores Feeder: geared container ship, 130 TEU, battery-electric with swappable battery containers, two island loops plus a transit each week
Specialization 1	SP01 · Longitudinal Strength Calculation
Specialization 2	SP02 · Propulsion Matching
Deadline	Dec 13, 2026, one ZIP file by e-mail to luis@veranavis.com
ZIP name	Group1_Project.zip
Presentation	Dec 18, 2026, 15 minutes per group

How the work is split. The in-class exercises are individual practice of each method on the reference ship (the 37,600 t oil/chemical tanker) and are not graded. This project is your group's work, done outside class: you apply each method to the Azores Feeder and complete the full scope of every deliverable below. A report that only repeats the class exercises does not meet the scope.

Planning (advised). A work plan is not a deliverable, but plan internally: a rough approach and a split of the tasks across the group, so that the work, the presentation and the knowledge of the whole project are shared evenly by every member. You are welcome to share the plan and discuss it with the instructor.

Code flowchart (mandatory). Include a flowchart of your code that shows how the information flows: inputs, each calculation step, the iterations and the outputs. Use the symbols of ISO 5807, the standard for data, program and system flowcharts.

Common deliverables D1–D10

#	Deliverable	Practised in class (individual, reference ship)	Your project must deliver (group, outside class)
D1	Design Basis	Ex. 1.2 does the owner's brief add up, classify the ship · Ex. 1.3 design phases and documents	130 TEU at 10 t per TEU (1,300 t cargo) · battery containers as extra deadweight, mass from supplier data · 11–12 kn · two loops plus the transit in a week · limits at all eight ports, each with its source · every requirement tagged Given / Assumed / To be checked
D2	Reference Ship Database	Ex. 1.4 first Δ, L, B, T, D from the App. A regressions, with 95% bands	At least 8 small geared container ships and battery ships · other families as your chosen hull needs · sources for every ship · your own regressions and bounds
D3	Ship Model	Ex. 1.5 displacement equation · Ex. 2.3 main dimensions · Ex. 2.4 power · Ex. 3.1 weights · Ex. 3.2 capacity · Ex. 3.3 engine and propeller	Volume-carrier sequence · inputs, outputs and calculation sequence · open-water propeller efficiency η0 as a stated, justified assumption (0.60) · electric drive, two shafts · batteries as deadweight · crane in the lightship · closed weight balance
D4	Voyage and Financial Model	Ex. 3.4 round voyage and fuel · Ex. 4.1 building cost · Ex. 4.2 RFR and NPV	Both loops and the transit: energy per leg, battery swaps and charging, electricity and port costs · building cost, financing, RFR and NPV · ship model outputs as input

#	Deliverable	Practised in class (individual, reference ship)	Your project must deliver (group, outside class)
D5	Validation Note	No class exercise: done by the group	Your model against the reference families you chose, with the choice justified · error per quantity · model corrected where needed
D6	Optimization Report	Ex. 5.1 synthesis model · Ex. 5.2 optimization (demonstration + individual step) · Ex. 5.3 multi-objective trade-off	Minimum battery capacity that completes both loops in one week · justified objective · active constraints · optimum against the requirements
D7	Compliance Sheet	Ex. 4.3 GM, freeboard and attained EEDI	EEDI and CII: show why they do not apply, and report energy per TEU-mile · battery rules (BV) · stability while the crane lifts · freeboard · draught at every port
D8	General Arrangement and Hull Form	Ex. 5.4 sectional area curve and LCB · Ex. 6.1 sections, bow and stern · Ex. 6.2 GA and capacity plan	Deck container layout and crane reach · battery slots aft · hull form of your choice, justified (a PSV-type hull is only a suggestion) · A3 sketch (profile and midship section) · hydrostatics check
D9	Specialization Documents (2)	No class exercise: self-study decks	One calculation note per specialization, with its required drawings (see Your specializations)
D10	Datasheet and Presentation	No class exercise	One-page A4 datasheet (0.4 Datasheet Template) · report on the 0.3 Report Template · presentation of 15 minutes

Exercises 7.1 and 7.2 (digital transition and AI) come after the deadline: your group may show them in the presentation, but they are not part of the ZIP.